

stat426exam

SM

2026-05-10

```
# Data
tab <- matrix(c(42, 18,
               30, 30),
              nrow = 2, byrow = TRUE)

rownames(tab) <- c("A", "B")
colnames(tab) <- c("Improved", "Not_improved")

tab

##      Improved Not_improved
## A          42           18
## B          30           30

#a.
# Let  $Y_A \sim \text{Binomial}(n_A = 60, p_A)$ , where  $p_A$  is the probability of improving for treatment A.
# Let  $Y_B \sim \text{Binomial}(n_B = 60, p_B)$  if  $p_B$  is the probability of improvement for treatment B.
# Here, the row totals are at 60 patients per treatment group. A multinomial sampling model would have the total sample size at  $N = 120$ , but the counts in the different cells would be treated as outcomes from one multinomial distribution.

#b.

a <- 42 # A improved
b <- 18 # A not improved
c <- 30 # B improved
d <- 30 # B not improved

nA <- a + b
nB <- c + d

pA <- a / nA
pB <- c / nB

RR <- pA / pB

SE_log_RR <- sqrt((1/a) - (1/nA) + (1/c) - (1/nB))

CI_RR <- exp(log(RR) + c(-1, 1) * 1.96 * SE_log_RR)
```

```

RR
## [1] 1.4

CI_RR
## [1] 1.034624 1.894409

# The estimated improvement risk for treatment A is 0.70. The estimated improvement risk for treatment B is 0.50.
#
#  $RR = 0.70 / 0.50 = 1.40$ 
#
# The approximate 95% confidence interval for the relative risk is:
#
# (1.0346, 1.8944)
#
# That means that having treatment A puts you at a 1.4 times chance of improving compared to group B. The confidence interval is entirely above 1, which means there is a significantly higher chance of improving with treatment A compared to B.

#c.

OR <- (a * d) / (b * c)
OR
## [1] 2.333333

# Odds of improvement for A = 42/18
# Odds of improvement for B = 30/30
# Odds ratio =  $(42*30)/(18*30) = 2.333$ 
#
# The odds of improvement are about 2.33 times higher for treatment A than for treatment B.

# d.
chisq_result <- chisq.test(tab, correct = FALSE)

chisq_result$expected

##      Improved Not_improved
## A           36             24
## B           36             24

chisq_result$statistic

```

```
## X-squared
##          5

chisq_result$parameter

## df
##  1

chisq_result$p.value

## [1] 0.02534732

# H0: Treatment and improvement outcome are independent.
# Ha: Treatment and improvement outcome are associated.

# Expected counts:
# A improved = 60*72/120 = 36
# A not improved = 60*48/120 = 24
# B improved = 60*72/120 = 36
# B not improved = 60*48/120 = 24
#
# The Pearson chi-square statistic is 5.00 with 1 degree of freedom. The p-value is 0.02535. Since the p-value is less than 0.05, we reject the null hypothesis. There is statistically significant evidence of an association between treatment and improvement outcome.

# e.

df <- as.data.frame(as.table(tab))
names(df) <- c("Treatment", "Outcome", "Count")

df

##   Treatment      Outcome Count
## 1         A      Improved    42
## 2         B      Improved    30
## 3         A Not_improved    18
## 4         B Not_improved    30

fit_ind <- glm(Count ~ Treatment + Outcome,
               family = poisson,
               data = df)

fit_sat <- glm(Count ~ Treatment * Outcome,
               family = poisson,
               data = df)

lrt <- anova(fit_ind, fit_sat, test = "Chisq")
lrt
```

```
## Analysis of Deviance Table
##
## Model 1: Count ~ Treatment + Outcome
## Model 2: Count ~ Treatment * Outcome
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1          1      5.0414
## 2          0      0.0000  1   5.0414  0.02475 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# The independence loglinear model is:
#  $\log(\mu_{ij}) = \lambda + \lambda_i^{\text{Treatment}} + \lambda_j^{\text{Outcome}}$ 
#
# The saturated model w/ interactions
#  $\log(\mu_{ij}) = \lambda + \lambda_i^{\text{Treatment}} + \lambda_j^{\text{Outcome}}$ 
#           +  $\lambda_{ij}^{\text{(Treatment*Outcome)}}$ 
#
# Testing independence is the same as testing interaction terms needed. The likelihood-ratio statistic is 5.0414 with 1 degree of freedom. The p-value is 0.02475. Since the p-value is less than 0.05, we reject the null hypothesis of independence. This means treatment and outcome interaction terms are needed.
# This test is related to the Pearson chi-square test in d because both test independence for a 2 by 2 table. The Pearson test uses the sum of (observed - expected)2 / expected, while the likelihood-ratio test uses the difference in model deviations.

# f.

logit_data <- data.frame(
  X = c(rep(1, 60), rep(0, 60)),
  Y = c(rep(1, 42), rep(0, 18),
        rep(1, 30), rep(0, 30))
)

fit_logit <- glm(Y ~ X, family = binomial, data = logit_data)

summary(fit_logit)

##
## Call:
## glm(formula = Y ~ X, family = binomial, data = logit_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) 6.762e-16  2.582e-01   0.000   1.0000
## X           8.473e-01  3.821e-01   2.217   0.0266 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```

## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 161.52 on 119 degrees of freedom
## Residual deviance: 156.48 on 118 degrees of freedom
## AIC: 160.48
##
## Number of Fisher Scoring iterations: 4

coef(fit_logit)

## (Intercept)          X
## 6.762473e-16 8.472979e-01

exp(coef(fit_logit))

## (Intercept)          X
## 1.000000 2.333333

# Logit[P(Y = Improved | X)] = beta0 + beta1*X
# where X = 1 for treatment A and X = 0 for treatment B.
# The fitted coefficient for X is 0.8473. exp(0.8473) = 2.3333
# This is the same odds ratio from c. The fitted coefficient for X is the Log
# odds ratio comparing treatment A to treatment B. The coefficient is positive,
# so treatment A has higher fitted odds of improvement than treatment B.

dat <- data.frame(
  year = as.numeric(time(discoveries)),
  count = as.numeric(discoveries)
)

dat$period <- cut(
  dat$year,
  breaks = c(1859, 1889, 1919, 1949, 1959),
  labels = c("1860-1889", "1890-1919", "1920-1949", "1950-1959")
)

dat$cyear <- dat$year - mean(dat$year)

#a.

# Let Y_i be the number of discoveries in year i.
# Assume:
#
# Y_i ~ Poisson(mu_i)
#
# with model:
#
# log(mu_i) = beta0 + beta1*cyear_i
# Random
# Y_i follows a Poisson distribution.
# Systemic

```

```

# beta0 + beta1*cyear_i
#
# Link function: Log link.
# The log link is natural for count data because it make sure positive fitted
# means and models multiplicative changes in the expected count.

# b.

m1 <- glm(count ~ cyear, family = poisson(link = "log"), data = dat)

summary(m1)

##
## Call:
## glm(formula = count ~ cyear, family = poisson(link = "log"),
##      data = dat)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.119460   0.057475  19.477 < 2e-16 ***
## cyear       -0.005360   0.001982  -2.705  0.00683 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 164.68  on 99  degrees of freedom
## Residual deviance: 157.32  on 98  degrees of freedom
## AIC: 430.32
##
## Number of Fisher Scoring iterations: 5

drop1(m1, test = "Rao")

## Single term deletions
##
## Model:
## count ~ cyear
##      Df Deviance    AIC Rao score Pr(>Chi)
## <none>      157.32 430.32
## cyear   1    164.69 435.69    7.3513 0.006702 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# H0: beta1 = 0
# Ha: beta1 is not equal to 0
#
# The Rao score test statistic is 7.3513 with 1 degree of freedom. The p-value is 0.006702.

```

```

#
# Since the p-value is less than 0.05, we reject the null hypothesis. There is
# significant evidence of there being a time effect in the number of discoveries
# The Wald test from the model summary gives a similar conclusion  $z = -2.705$ 
# and  $p\text{-value} = 0.00683$ .

# c.

pearson_disp <- sum(residuals(m1, type = "pearson")^2) / m1$df.residual
deviance_disp <- deviance(m1) / m1$df.residual

pearson_disp
## [1] 1.541413

deviance_disp
## [1] 1.605264

m1_quasi <- glm(count ~ cyear, family = quasipoisson(link = "log"), data = dat)

summary(m1_quasi)
##
## Call:
## glm(formula = count ~ cyear, family = quasipoisson(link = "log"),
##      data = dat)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.11946    0.07136  15.688  <2e-16 ***
## cyear        -0.00536    0.00246  -2.179   0.0318 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.541413)
##
##      Null deviance: 164.68  on 99  degrees of freedom
## Residual deviance: 157.32  on 98  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5

# The estimated dispersion parameter using Pearson residuals is: 1.5414. The
# deviance-based dispersion estimate is: 1.6053 Since both estimates are greater
# than 1, there is evidence of overdispersion in the Poisson model.
# This does not substantially change the coefficient estimates, but it increases
# the standard errors. In the quasipoisson model, the coefficient estimate f

```

or cyear remains -0.00536 , but the standard error increases from 0.001982 to 0.00246 . As a result, the p -value increases from 0.00683 to 0.0318 which means that the original Poisson model underestimated variability and overstated the statistical significance.

d.

```
m2 <- glm(count ~ period, family = poisson(link = "log"), data = dat)
```

```
summary(m2)
```

```
##
## Call:
## glm(formula = count ~ period, family = poisson(link = "log"),
##      data = dat)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.23353    0.09853   12.519  < 2e-16 ***
## period1890-1919  0.09266    0.13623    0.680  0.496396
## period1920-1949 -0.22801    0.14800   -1.541  0.123411
## period1950-1959 -1.05121    0.30503   -3.446  0.000568 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 164.68  on 99  degrees of freedom
## Residual deviance: 143.05  on 96  degrees of freedom
## AIC: 420.05
##
## Number of Fisher Scoring iterations: 5
```

```
AIC(m1, m2)
```

```
##      df      AIC
## m1  2 430.3225
## m2  4 420.0528
```

The fitted coefficients show:

#

period1890-1919: 0.09266

period1920-1949: -0.22801

period1950-1959: -1.05121

Based on the baseline period 1860-1889, the period from 1950-1959 has significantly lower expected discovery counts.

AIC(m1) = 430.32


```

# AIC(m2) = 420.05
#
# Since the period model has the smaller AIC, we would prefer it.
# AIC balances model fit and model complexity by rewarding better fit while also penalizing for complexity of the model.
#
# The centered-year model assumes that there is a linear trend over time, while the period model allows different mean levels for the different periods in time.

# e.

lev <- hatvalues(m2)

lev
##           1           2           3           4           5           6
7
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##           8           9          10          11          12          13
14
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          15          16          17          18          19          20
21
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          22          23          24          25          26          27
28
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          29          30          31          32          33          34
35
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          36          37          38          39          40          41
42
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          43          44          45          46          47          48
49
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          50          51          52          53          54          55
56
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          57          58          59          60          61          62
63

```

```

## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          64          65          66          67          68          69
70
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          71          72          73          74          75          76
77
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          78          79          80          81          82          83
84
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.033333
33
##          85          86          87          88          89          90
91
## 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.03333333 0.100000
00
##          92          93          94          95          96          97
98
## 0.10000000 0.10000000 0.10000000 0.10000000 0.10000000 0.10000000 0.100000
00
##          99          100
## 0.10000000 0.10000000

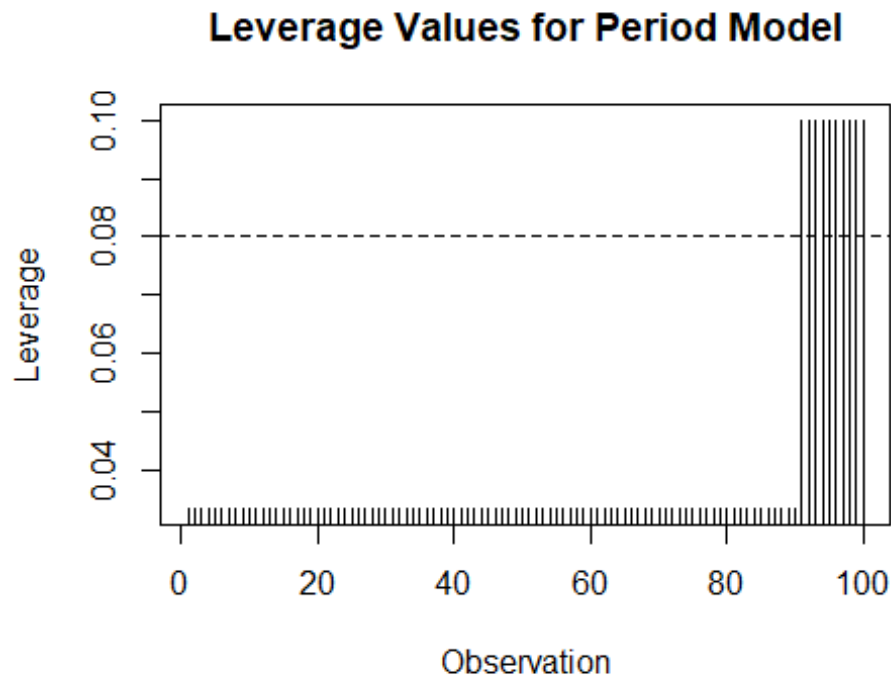
max(lev)

## [1] 0.1

plot(lev,
     type = "h",
     main = "Leverage Values for Period Model",
     xlab = "Observation",
     ylab = "Leverage")

abline(h = 2 * mean(lev), lty = 2)
abline(h = 3 * mean(lev), lty = 3)

```



```
dat[order(lev, decreasing = TRUE), ]
```

##	year	count	period	cyear
## 91	1950	2	1950-1959	40.5
## 92	1951	1	1950-1959	41.5
## 93	1952	4	1950-1959	42.5
## 94	1953	1	1950-1959	43.5
## 95	1954	1	1950-1959	44.5
## 96	1955	1	1950-1959	45.5
## 97	1956	0	1950-1959	46.5
## 98	1957	0	1950-1959	47.5
## 99	1958	2	1950-1959	48.5
## 100	1959	0	1950-1959	49.5
## 2	1861	3	1860-1889	-48.5
## 5	1864	0	1860-1889	-45.5
## 6	1865	3	1860-1889	-44.5
## 7	1866	2	1860-1889	-43.5
## 8	1867	3	1860-1889	-42.5
## 9	1868	6	1860-1889	-41.5
## 10	1869	1	1860-1889	-40.5
## 11	1870	2	1860-1889	-39.5
## 12	1871	1	1860-1889	-38.5
## 13	1872	2	1860-1889	-37.5
## 14	1873	1	1860-1889	-36.5
## 15	1874	3	1860-1889	-35.5
## 16	1875	3	1860-1889	-34.5
## 17	1876	3	1860-1889	-33.5

## 18	1877	5	1860-1889	-32.5
## 19	1878	2	1860-1889	-31.5
## 20	1879	4	1860-1889	-30.5
## 21	1880	4	1860-1889	-29.5
## 22	1881	0	1860-1889	-28.5
## 23	1882	2	1860-1889	-27.5
## 24	1883	3	1860-1889	-26.5
## 25	1884	7	1860-1889	-25.5
## 26	1885	12	1860-1889	-24.5
## 27	1886	3	1860-1889	-23.5
## 28	1887	10	1860-1889	-22.5
## 29	1888	9	1860-1889	-21.5
## 30	1889	2	1860-1889	-20.5
## 4	1863	2	1860-1889	-46.5
## 3	1862	0	1860-1889	-47.5
## 31	1890	3	1890-1919	-19.5
## 32	1891	7	1890-1919	-18.5
## 33	1892	7	1890-1919	-17.5
## 34	1893	2	1890-1919	-16.5
## 35	1894	3	1890-1919	-15.5
## 36	1895	3	1890-1919	-14.5
## 37	1896	6	1890-1919	-13.5
## 38	1897	2	1890-1919	-12.5
## 39	1898	4	1890-1919	-11.5
## 40	1899	3	1890-1919	-10.5
## 41	1900	5	1890-1919	-9.5
## 42	1901	2	1890-1919	-8.5
## 43	1902	2	1890-1919	-7.5
## 44	1903	4	1890-1919	-6.5
## 45	1904	0	1890-1919	-5.5
## 46	1905	4	1890-1919	-4.5
## 47	1906	2	1890-1919	-3.5
## 48	1907	5	1890-1919	-2.5
## 49	1908	2	1890-1919	-1.5
## 50	1909	3	1890-1919	-0.5
## 51	1910	3	1890-1919	0.5
## 52	1911	6	1890-1919	1.5
## 53	1912	5	1890-1919	2.5
## 54	1913	8	1890-1919	3.5
## 55	1914	3	1890-1919	4.5
## 56	1915	6	1890-1919	5.5
## 57	1916	6	1890-1919	6.5
## 58	1917	0	1890-1919	7.5
## 59	1918	5	1890-1919	8.5
## 60	1919	2	1890-1919	9.5
## 61	1920	2	1920-1949	10.5
## 62	1921	2	1920-1949	11.5
## 63	1922	6	1920-1949	12.5
## 64	1923	3	1920-1949	13.5
## 65	1924	4	1920-1949	14.5

```
## 66 1925      4 1920-1949 15.5
## 67 1926      2 1920-1949 16.5
## 68 1927      2 1920-1949 17.5
## 69 1928      4 1920-1949 18.5
## 70 1929      7 1920-1949 19.5
## 71 1930      5 1920-1949 20.5
## 72 1931      3 1920-1949 21.5
## 73 1932      3 1920-1949 22.5
## 74 1933      0 1920-1949 23.5
## 75 1934      2 1920-1949 24.5
## 76 1935      2 1920-1949 25.5
## 77 1936      2 1920-1949 26.5
## 78 1937      1 1920-1949 27.5
## 79 1938      3 1920-1949 28.5
## 80 1939      4 1920-1949 29.5
## 81 1940      2 1920-1949 30.5
## 82 1941      2 1920-1949 31.5
## 83 1942      1 1920-1949 32.5
## 84 1943      1 1920-1949 33.5
## 85 1944      1 1920-1949 34.5
## 86 1945      2 1920-1949 35.5
## 87 1946      1 1920-1949 36.5
## 88 1947      4 1920-1949 37.5
## 89 1948      4 1920-1949 38.5
## 90 1949      3 1920-1949 39.5
## 1  1860      5 1860-1889 -49.5
```

In the period model, most observations have Leverage equal to 0.0333, while observations in the final period have Leverage equal to 0.10. Since smaller groups seem to produce larger Leverage values, the fitted model is somewhat more sensitive to observations in the final period.

#

However, the Leverage values are not extremely large, so there is not strong evidence that any single observation too influential.

```
b0 <- -2.20
```

```
b1 <- -0.50
```

```
b2 <- 0.35
```

```
b3 <- -0.12
```

```
se_b3 <- 0.06
```

a.

For a fixed value the model is $\text{Logit}[P(Y = 1 \mid A, X)] = b_0 + b_1 A + b_2 X + b_3 A X$

#

Treatment-control odds ratio is $\exp(b_1 + b_3 x)$

At $X = 0$: $OR = \exp(-0.50 + -0.12 \cdot 0) = \exp(-0.50) = 0.6065$

```

# At X = 5: OR = exp(-0.50 + -0.12*5) = exp(-1.10) = 0.3329
# The treatment-control odds ratio decreases as baseline severity increases.
# Since both odds ratios are below 1, the treatment is associated with lower odds of an adverse event compared with control, and this protective effect is stronger at X = 5 than at X = 0.

# At X = 0, OR = exp(-0.50) = 0.607. At X = 5, OR = exp(-0.50 + -0.12*5) = exp(-1.10) = 0.333. The treatment-control odds ratio decreases as X increases.

# b.

z <- b3 / se_b3
p_value <- 2 * (1 - pnorm(abs(z)))

z
## [1] -2

p_value
## [1] 0.04550026

# b3 = -0.12 means that each 1-unit increase in baseline severity X multiplies the treatment-control odds ratio by exp(-0.12).
exp(b3)
## [1] 0.8869204

# exp(-0.12) = 0.887.
# So for each 1-unit increase in X, the treatment-control odds ratio is multiplied by about 0.887, or decreases by about 11.3%.

# H0: b3 = 0
# Ha: b3 is not 0
#
# z = -0.12 / 0.06 = -2.00
# p-value = about 0.046
#
# Since p < 0.05, reject H0. There is evidence that the treatment effect changes with baseline severity.

# Likelihood-ratio test explanation:
# Full model: Y ~ A + X + A:X
# Reduced model: Y ~ A + X
#
#
# The p-value comes from a chi-square test with 1 degree of freedom.

```

```

# c.
eta_treated <- b0 + b1*1 + b2*5 + b3*1*5
eta_control <- b0 + b1*0 + b2*5 + b3*0*5

p_treated <- exp(eta_treated) / (1 + exp(eta_treated))
p_control <- exp(eta_control) / (1 + exp(eta_control))

p_treated
## [1] 0.1750863

p_control
## [1] 0.3893608

diff_probs <- p_treated - p_control
diff_probs
## [1] -0.2142745

# For treated patients at X = 5: eta = -2.20 - 0.50 + 0.35(5) - 0.12(5) = -1.55
# Probability = 0.175.
# For control patients at X = 5: eta = -2.20 + 0.35(5) = -0.45
# Probability = 0.389.
#
# Difference in probabilities = 0.175 - 0.389 = -0.214.
# Odds ratios compare odds, while differences in proportions compare actual probabilities. Odds ratios can sound large or small depending on the baseline risk, while probability differences are often easier to interpret directly.

# d.

# Linear predictor for treated patient with X = 5: eta = b0 + b1 + 5*b2 + 5*b3

eta_hat <- b0 + b1 + 5*b2 + 5*b3
p_hat <- plogis(eta_hat)

eta_hat
## [1] -1.55

p_hat
## [1] 0.1750863

# To construct the 95% CI:
# 1. Use the coefficient vector: c = (1, 1, 5, 5)
# 2. Compute: eta_hat = b0 + b1 + 5*b2 + 5*b3
# 3. Compute the standard error: SE(eta_hat) = sqrt(c' V c) where V is the covariance matrix of the coefficient estimates.

```

*# 4. Construct the CI on the Logit scale: $\eta_{\text{hat}} \pm 1.96 * SE(\eta_{\text{hat}})$*
5. Convert endpoints back to probability scale using: $p = \exp(\eta) / (1 + \exp(\eta))$
Since only individual standard errors are given, not the full covariance matrix, the exact CI cannot be computed from the table alone.

e.

If treatment has levels A, B, and C, with A as the reference group, use two indicator variables:

B_i = 1 if treatment is B, 0 otherwise

C_i = 1 if treatment is C, 0 otherwise

#

*# Additive Logistic model: $\text{Logit}[P(Y=1)] = b_0 + b_B * B_i + b_C * C_i + b_X * X$*

In this model: bB compares B to A. bC compares C to A. To get the odds ratio comparing C with B: $OR_{C_vs_B} = \exp(b_C - b_B)$

data(esoph)

esoph2 <- esoph

esoph2\$high_alc <- ifelse(esoph2\$alcgp == "80-119" | esoph2\$alcgp == "120+",
"High alcohol",
"Lower alcohol")

a.

marginal_tab <- aggregate(cbind(ncases, ncontrols) ~ high_alc,
data = esoph2,
sum)

marginal_tab

```
##           high_alc ncases ncontrols
```

```
## 1 High alcohol      96         109
```

```
## 2 Lower alcohol    104         666
```

tab2 <- as.matrix(marginal_tab[, c("ncases", "ncontrols")])

rownames(tab2) <- marginal_tab\$high_alc

tab2

```
##           ncases ncontrols
```

```
## High alcohol      96         109
```

```
## Lower alcohol    104         666
```

cases_high <- tab2["High alcohol", "ncases"]

controls_high <- tab2["High alcohol", "ncontrols"]

cases_low <- tab2["Lower alcohol", "ncases"]

controls_low <- tab2["Lower alcohol", "ncontrols"]


```
marginal_OR <- (cases_high * controls_low) / (controls_high * cases_low)
marginal_OR
```

```
## [1] 5.640085
```

After collapsing over age and tobacco group, the marginal 2 by 2 table gives 96 cases and 109 controls in the high alcohol group, and 104 cases and 666 controls in the lower alcohol group. The estimated marginal odds ratio is 5.64. This means the marginal odds of being a cancer case are about 5.64 times higher for the high alcohol group compared with the lower alcohol group.

b.

```
age_tab <- aggregate(cbind(ncases, ncontrols) ~ agegp + high_alc,
                     data = esoph2,
                     sum)
```

```
age_tab
```

```
##      agegp      high_alc ncases ncontrols
## 1  25-34 High alcohol      1         9
## 2  35-44 High alcohol      4        26
## 3  45-54 High alcohol     25        29
## 4  55-64 High alcohol     42        27
## 5  65-74 High alcohol     19        18
## 6   75+ High alcohol      5         0
## 7  25-34 Lower alcohol      0       106
## 8  35-44 Lower alcohol      5       164
## 9  45-54 Lower alcohol     21       138
## 10 55-64 Lower alcohol     34       139
## 11 65-74 Lower alcohol     36        88
## 12  75+ Lower alcohol      8        31
```

```
age_ORs <- do.call(rbind, lapply(split(age_tab, age_tab$agegp), function(d) {
  high <- d[d$high_alc == "High alcohol", ]
  low <- d[d$high_alc == "Lower alcohol", ]

  OR <- (high$ncases * low$ncontrols) / (high$ncontrols * low$ncases)

  data.frame(agegp = unique(d$agegp),
             OR = OR)
})))
```

```
age_ORs
```

```
##      agegp      OR
## 25-34 25-34 Inf
```

```
## 35-44 35-44 5.046154
## 45-54 45-54 5.665025
## 55-64 55-64 6.359477
## 65-74 65-74 2.580247
## 75+      75+      Inf
```

The age-specific odds ratios compare high alcohol consumption with lower alcohol consumption within each age group.
Overall, the age-specific results show that high alcohol consumption is associated with higher odds of cancer within the age groups. However, the odds ratios are not perfectly homogeneous because some age groups have sparse or zero cells, especially the youngest and oldest age groups. The marginal odds ratio of 5.64 is similar in direction to the age-adjusted results, so there is not strong evidence of Simpson's paradox.

```
# c.
```

```
age_levels <- levels(esoph2$agegp)
```

```
cmh_array <- array(0,
  dim = c(2, 2, length(age_levels)),
  dimnames = list(
    Alcohol = c("High alcohol", "Lower alcohol"),
    Status = c("Case", "Control"),
    Age = age_levels
  ))
```

```
for (ag in age_levels) {
```

```
  temp <- subset(esoph2, agegp == ag)
  temp2 <- aggregate(cbind(ncases, ncontrols) ~ high_alc,
    data = temp,
    sum)
```

```
  for (i in 1:nrow(temp2)) {
    cmh_array[temp2$high_alc[i], "Case", ag] <- temp2$ncases[i]
    cmh_array[temp2$high_alc[i], "Control", ag] <- temp2$ncontrols[i]
  }
}
```

```
cmh_array
```

```
## , , Age = 25-34
##
##           Status
## Alcohol    Case Control
## High alcohol    1      9
## Lower alcohol   0     106
```

```

##
## , , Age = 35-44
##
##           Status
## Alcohol   Case Control
##   High alcohol    4      26
##   Lower alcohol   5     164
##
## , , Age = 45-54
##
##           Status
## Alcohol   Case Control
##   High alcohol   25      29
##   Lower alcohol  21     138
##
## , , Age = 55-64
##
##           Status
## Alcohol   Case Control
##   High alcohol   42      27
##   Lower alcohol  34     139
##
## , , Age = 65-74
##
##           Status
## Alcohol   Case Control
##   High alcohol   19      18
##   Lower alcohol  36      88
##
## , , Age = 75+
##
##           Status
## Alcohol   Case Control
##   High alcohol    5       0
##   Lower alcohol    8      31

mantelhaen.test(cmh_array)

##
## Mantel-Haenszel chi-squared test with continuity correction
##
## data:  cmh_array
## Mantel-Haenszel X-squared = 83.215, df = 1, p-value < 2.2e-16
## alternative hypothesis: true common odds ratio is not equal to 1
## 95 percent confidence interval:
##  3.562131 7.467743
## sample estimates:
## common odds ratio
##      5.157623

```

H0: After adjusting for age group, alcohol group and cancer status are conditionally independent. Ha: After adjusting for age group, alcohol group and cancer status are associated.

#The Cochran-Mantel-Haenszel test statistic is 83.215 with 1 degree of freedom, and the p-value is less than 2.2e-16. Since the p-value is extremely small, we reject the null hypothesis. There is very strong evidence that alcohol consumption is associated with cancer status after adjusting for age group. The estimated common odds ratio is 5.16, with a 95% confidence interval of 3.56 to 7.47.

d.

```
fit_common <- glm(cbind(ncases, ncontrols) ~ high_alc + agegp,
                  family = binomial,
                  data = esoph2)
```

```
summary(fit_common)
```

```
##
## Call:
## glm(formula = cbind(ncases, ncontrols) ~ high_alc + agegp, family = binomial,
##      data = esoph2)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.65385     0.23157  -2.824   0.00475 **
## high_alcLower alcohol -1.66989     0.18960  -8.807   < 2e-16 ***
## agegp.L         3.29876     0.65272   5.054 4.33e-07 ***
## agegp.Q        -1.45630     0.59182  -2.461   0.01387 *
## agegp.C         0.05813     0.45110   0.129   0.89746
## agegp^4         0.23796     0.31131   0.764   0.44465
## agegp^5        -0.18983     0.20187  -0.940   0.34703
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 367.95  on 87  degrees of freedom
## Residual deviance: 167.39  on 81  degrees of freedom
## AIC: 296.44
##
## Number of Fisher Scoring iterations: 6
```

```
exp(coef(fit_common))
```

```
##              (Intercept) high_alcLower alcohol              agegp.L
##              0.5200372          0.1882678          27.0791541
##              agegp.Q              agegp.C              agegp^4
##              0.2330965          1.0598546          1.2686585
```

```
##          agegp^5
##          0.8270994
```

```
exp(confint(fit_common))
```

```
## Waiting for profiling to be done...
```

```
##              2.5 %      97.5 %
## (Intercept)    0.30006381  0.7874295
## high_alcLower alcohol 0.12940697  0.2723310
## agegp.L        9.42772950 158.2948961
## agegp.Q        0.04668086  0.6014749
## agegp.C        0.48384704  3.3358571
## agegp^4        0.64124767  2.3119830
## agegp^5        0.55503270  1.2311427
```

The logistic regression model used cancer case-control status as the response and included alcohol group and age group as predictors. The coefficient for Lower alcohol is -1.6699, so the adjusted odds ratio for Lower alcohol compared with high alcohol is $\exp(-1.6699) = 0.188$. Equivalently, the adjusted odds ratio for high alcohol compared with lower alcohol is $1 / 0.188 = 5.31$. This means that, after adjusting for age group, the odds of being a cancer case are about 5.31 times higher for the high alcohol group compared with the lower alcohol group.

e.

```
deviance(fit_common)
```

```
## [1] 167.3869
```

```
df.residual(fit_common)
```

```
## [1] 81
```

```
gof_p <- 1 - pchisq(deviance(fit_common), df.residual(fit_common))
gof_p
```

```
## [1] 5.73755e-08
```

To test whether the model fits well, I used the residual deviance goodness-of-fit test.

H0: The model fits adequately.

Ha: The model does not fit adequately.

The residual deviance is 167.39 on 81 degrees of freedom, with a p-value of about $5.74e-08$.

Since this p-value is very small, we reject the null hypothesis. This suggests that the model with only alcohol group and age group does not fit the data well. There is evidence of lack of fit, meaning the model may be missing an important structure such as an interaction or another predictor.

f.

```

fit_interaction <- glm(cbind(ncases, ncontrols) ~ high_alc * agegp,
                      family = binomial,
                      data = esoph2)

summary(fit_interaction)

##
## Call:
## glm(formula = cbind(ncases, ncontrols) ~ high_alc * agegp, family = binomial,
##      data = esoph2)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.4923    513.5373   0.005   0.996
## high_alcLower alcohol      -7.3563    564.1746  -0.013   0.990
## agegp.L           13.2350   1841.3833   0.007   0.994
## agegp.Q            9.0600   1680.9453   0.005   0.996
## agegp.C            6.5980   1148.3043   0.006   0.995
## agegp^4            4.2556    582.2967   0.007   0.994
## agegp^5            1.0801    194.0991   0.006   0.996
## high_alcLower alcohol:agegp.L  -1.0122   2022.9527  -0.001   1.000
## high_alcLower alcohol:agegp.Q -18.8794   1846.6947  -0.010   0.992
## high_alcLower alcohol:agegp.C  -1.0881   1261.5327  -0.001   0.999
## high_alcLower alcohol:agegp^4  -7.0785    639.7140  -0.011   0.991
## high_alcLower alcohol:agegp^5  -0.4147    213.2383  -0.002   0.998
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 367.95  on 87  degrees of freedom
## Residual deviance: 156.35  on 76  degrees of freedom
## AIC: 295.4
##
## Number of Fisher Scoring iterations: 17

anova(fit_common, fit_interaction, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: cbind(ncases, ncontrols) ~ high_alc + agegp
## Model 2: cbind(ncases, ncontrols) ~ high_alc * agegp
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         81      167.39
## 2         76      156.35  5   11.041  0.05057 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# The larger model included an interaction between alcohol group and age group. The likelihood-ratio test compares the common-odds model to the interaction model.

```

H0: There is no alcohol-by-age interaction. This means the alcohol odds ratio is homogeneous across age groups.
 # Ha: There is an alcohol-by-age interaction. This means the alcohol effect differs by age group.
 # The likelihood-ratio test statistic is 11.041 with 5 degrees of freedom, and the p-value is 0.05057. This means at the 5% level we fail to reject the null hypothesis. There is evidence that the alcohol effect varies by age group.
 Overall, the common odds ratio model is still reasonable, but the interaction may be worth considering.

g.

```
fitted_values <- fitted(fit_common)
```

```
range(fitted_values)
```

```
## [1] 0.006341062 0.641625004
```

```
fitted_values
```

```
##          1          2          3          4          5          6
## 0.006341062 0.006341062 0.006341062 0.006341062 0.006341062 0.006341062
##          7          8          9         10         11         12
## 0.006341062 0.006341062 0.032784745 0.032784745 0.032784745 0.032784745
##         13         14         15         16         17         18
## 0.032784745 0.032784745 0.032784745 0.028971200 0.028971200 0.028971200
##         19         20         21         22         23         24
## 0.028971200 0.028971200 0.028971200 0.028971200 0.028971200 0.136795575
##         25         26         27         28         29         30
## 0.136795575 0.136795575 0.136795575 0.136795575 0.136795575 0.136795575
##         31         32         33         34         35         36
## 0.135218355 0.135218355 0.135218355 0.135218355 0.135218355 0.135218355
##         37         38         39         40         41         42
## 0.135218355 0.135218355 0.453708919 0.453708919 0.453708919 0.453708919
##         43         44         45         46         47         48
## 0.453708919 0.453708919 0.453708919 0.453708919 0.207369066 0.207369066
##         49         50         51         52         53         54
## 0.207369066 0.207369066 0.207369066 0.207369066 0.207369066 0.207369066
##         55         56         57         58         59         60
## 0.581523937 0.581523937 0.581523937 0.581523937 0.581523937 0.581523937
##         61         62         63         64         65         66
## 0.581523937 0.581523937 0.252095765 0.252095765 0.252095765 0.252095765
##         67         68         69         70         71         72
## 0.252095765 0.252095765 0.252095765 0.641625004 0.641625004 0.641625004
##         73         74         75         76         77         78
## 0.641625004 0.641625004 0.641625004 0.641625004 0.641625004 0.251212131
##         79         80         81         82         83         84
## 0.251212131 0.251212131 0.251212131 0.251212131 0.251212131 0.251212131
```

```
##           85           86           87           88
## 0.640545378 0.640545378 0.640545378 0.640545378
```

```
xtabs(ncases ~ high_alc + agegp, data = esoph2)
```

```
##           agegp
## high_alc    25-34 35-44 45-54 55-64 65-74 75+
## High alcohol      1      4     25     42     19      5
## Lower alcohol      0      5     21     34     36      8
```

```
xtabs(ncontrols ~ high_alc + agegp, data = esoph2)
```

```
##           agegp
## high_alc    25-34 35-44 45-54 55-64 65-74 75+
## High alcohol      9     26     29     27     18      0
## Lower alcohol   106    164    138    139     88    31
```

For the common-odds model in part d, there are not major signs of nonexistence of the MLE. The fitted probabilities range from about 0.0063 to 0.6416, so they are not exactly 0 or 1, and the coefficient estimates are not extremely large.

#

However, the interaction model from part f does show warning signs of quasi-complete separation. There are zero cells in the data, including 0 cases for the lower alcohol group in age group 25-34 and 0 controls for the high alcohol group in age group 75+. This explains why the interaction model produced really big standard errors and unstable coefficient estimates.

```
library(MASS)
library(nnet)
library(VGAM)
```

```
## Warning: package 'VGAM' was built under R version 4.5.3
```

```
## Loading required package: stats4
```

```
## Loading required package: splines
```

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.5.3
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 5.0
```

```
data(survey)
dat <- na.omit(survey[, c("Smoke", "Sex", "Age", "Exer")])
```

```
str(dat)
```

```
## 'data.frame':    235 obs. of  4 variables:
## $ Smoke: Factor w/ 4 levels "Heavy","Never",...: 2 4 3 2 2 2 2 2 2 2 ...
```



```
## $ Sex : Factor w/ 2 levels "Female","Male": 1 2 2 2 2 1 2 1 2 2 ...
## $ Age : num 18.2 17.6 16.9 20.3 23.7 ...
## $ Exer : Factor w/ 3 levels "Freq","None",...: 3 2 2 2 3 3 1 1 3 3 ...
## - attr(*, "na.action")= 'omit' Named int [1:2] 70 137
## ..- attr(*, "names")= chr [1:2] "70" "137"
```

```
summary(dat)
```

```
##      Smoke      Sex      Age      Exer
## Heavy: 11 Female:118 Min.   :16.75 Freq:114
## Never:188 Male   :117 1st Qu.:17.62 None: 23
## Occas: 19          Median :18.58 Some: 98
## Regul: 17          Mean   :20.38
##          3rd Qu.:20.17
##          Max.    :73.00
```

```
dat$Smoke <- factor(dat$Smoke,
                    levels = c("Never", "Occas", "Regul", "Heavy"))
```

```
# a.
```

```
fit_multinom <- multinom(Smoke ~ Sex + Age + Exer, data = dat)
```

```
## # weights: 24 (15 variable)
## initial value 325.779175
## iter 10 value 169.160932
## iter 20 value 163.162368
## final value 163.162163
## converged
```

```
summary(fit_multinom)
```

```
## Call:
## multinom(formula = Smoke ~ Sex + Age + Exer, data = dat)
##
## Coefficients:
##      (Intercept)      SexMale      Age      ExerNone      ExerSome
## Occas    -1.994620  0.07262113 -0.0007570218  0.1815997 -1.06367548
## Regul    -3.301476  0.98127650  0.0190961688 -0.6175943 -0.07236724
## Heavy    -3.049864  0.18205520  0.0211863741 -0.4041945 -0.79359238
##
## Std. Errors:
##      (Intercept)      SexMale      Age      ExerNone      ExerSome
## Occas    0.9167190  0.4903765  0.04039365  0.6979195  0.6024554
## Regul    0.8835780  0.5591027  0.03233434  1.0916612  0.5371679
## Heavy    0.9542312  0.6318929  0.03711244  1.1027195  0.7151235
##
## Residual Deviance: 326.3243
## AIC: 356.3243
```

```
zvals <- summary(fit_multinom)$coefficients /
          summary(fit_multinom)$standard.errors
```

```
zvals
```

```
##      (Intercept)  SexMale      Age  ExerNone  ExerSome
## Occas   -2.175825  0.1480926 -0.01874111  0.2602015 -1.765567
## Regul   -3.736486  1.7550917  0.59058479 -0.5657381 -0.134720
## Heavy   -3.196148  0.2881108  0.57086989 -0.3665433 -1.109728
```

```
pvals <- 2 * (1 - pnorm(abs(zvals)))
pvals
```

```
##      (Intercept)  SexMale      Age  ExerNone  ExerSome
## Occas  0.0295683449  0.88226970  0.9850476  0.7947084  0.0774685
## Regul  0.0001866101  0.07924362  0.5547987  0.5715719  0.8928333
## Heavy  0.0013927581  0.77326190  0.5680878  0.7139597  0.2671164
```

```
exp(coef(fit_multinom))
```

```
##      (Intercept)  SexMale      Age  ExerNone  ExerSome
## Occas  0.13606533  1.075323  0.9992433  1.1991341  0.3451848
## Regul  0.03682876  2.667860  1.0192797  0.5392401  0.9301892
## Heavy  0.04736538  1.199680  1.0214124  0.6675143  0.4522173
```

The baseline-category logit model treats Smoke as a nominal response, with Never as the baseline category. The model compares Occas, Regul, and Heavy smoking to Never smoking.

The fitted logits are:

#

$\log(P(\text{Occas}) / P(\text{Never})) =$

$-1.9946 + 0.0726 \cdot \text{SexMale} - 0.0008 \cdot \text{Age} + 0.1816 \cdot \text{ExerNone} - 1.0637 \cdot \text{ExerSome}$

#

$\log(P(\text{Regul}) / P(\text{Never})) =$

$-3.3015 + 0.9813 \cdot \text{SexMale} + 0.0191 \cdot \text{Age} - 0.6176 \cdot \text{ExerNone} - 0.0724 \cdot \text{ExerSome}$

$\log(P(\text{Heavy}) / P(\text{Never})) =$

$-3.0499 + 0.1821 \cdot \text{SexMale} + 0.0212 \cdot \text{Age} - 0.4042 \cdot \text{ExerNone} - 0.7936 \cdot \text{ExerSome}$

For Sex, the coefficient for males is positive in all three smoking categories. This means males have higher estimated odds than females of being Occas, Regul, or Heavy smokers compared with Never smokers.

#b.

```
dat$Smoke_ord <- ordered(dat$Smoke,
                          levels = c("Never", "Occas", "Regul", "Heavy"))
```

```
fit_polr <- polr(Smoke_ord ~ Sex + Age + Exer,
                 data = dat,
                 Hess = TRUE)
```

```
summary(fit_polr)
```

```
## Call:
## polr(formula = Smoke_ord ~ Sex + Age + Exer, data = dat, Hess = TRUE)
##
## Coefficients:
##              Value Std. Error t value
## SexMale      0.42576    0.33398  1.2748
## Age          0.01471    0.02291  0.6421
## ExerNone    -0.20244    0.54709 -0.3700
## ExerSome   -0.57876    0.36417 -1.5893
##
## Intercepts:
##              Value Std. Error t value
## Never|Occas  1.6823   0.5605    3.0013
## Occas|Regul  2.3080   0.5742    4.0196
## Regul|Heavy  3.3307   0.6223    5.3522
##
## Residual Deviance: 330.753
## AIC: 344.753
```

```
coefs <- coef(summary(fit_polr))
pvals_polr <- 2 * (1 - pnorm(abs(coefs[, "t value"])))
```

```
cbind(coefs, p_value = pvals_polr)
```

```
##              Value Std. Error    t value    p_value
## SexMale      0.42575979 0.33398073  1.2748035 2.023788e-01
## Age          0.01471303 0.02291328  0.6421178 5.207967e-01
## ExerNone    -0.20244262 0.54708592 -0.3700381 7.113541e-01
## ExerSome   -0.57875615 0.36416582 -1.5892654 1.120005e-01
## Never|Occas  1.68227161 0.56051884  3.0012758 2.688509e-03
## Occas|Regul  2.30803836 0.57419377  4.0196158 5.829313e-05
## Regul|Heavy  3.33071786 0.62230600  5.3522188 8.688225e-08
```

```
exp(coef(fit_polr))
```

```
## SexMale      Age ExerNone ExerSome
## 1.5307530 1.0148218 0.8167333 0.5605952
```

The proportional odds model treats Smoke as ordinal: Never < Occas < Regul < Heavy.

The cumulative-logit model is: $\log(P(\text{Smoke} \leq j) / P(\text{Smoke} > j)) = \alpha_j - \beta_1 \text{SexMale} - \beta_2 \text{Age} - \beta_3 \text{ExerNone} - \beta_4 \text{ExerSome}$

The fitted predictor coefficients are: SexMale = 0.4258, Age = 0.0147, Exer None = -0.2024, ExerSome = -0.5788

The odds ratio for SexMale is $\exp(0.4258) = 1.53$. This means males have about 1.53 times the odds of being in a higher smoking category than females holding Age and Exer constant. The p-value for SexMale is 0.202, so there is not strong evidence that Sex is associated with smoking level in the proportional odds model. The proportional odds assumption means that the effects of Sex, Age, and Exer are assumed to be the same across all cumulative splits of the smoking outcome.

#c.
The baseline-category logit model is more flexible because it allows each predictor to have a different effect for the different combinations. The proportional odds model is more restrictive because it treats smoke as ordinal with one common effect of each predictor across the logits.
The multinomial model predicts the Sex effect differs by category, with the strongest effect for Regular smoking. The proportional odds model summarizes this with one common Sex effect. Therefore, the multinomial model gives more detail, while the proportional odds model is simpler and easier to interpret if the proportional odds assumption is reasonable.

```
# d.
X <- model.matrix(~ (Sex + Age + Exer)^2, data = dat)[, -1]

y <- dat$Smoke

set.seed(123456789)

alpha_grid <- seq(0, 1, by = 0.1)

cv_errors <- numeric(length(alpha_grid))
cv_models <- vector("list", length(alpha_grid))

for(i in seq_along(alpha_grid)) {

  cvfit <- cv.glmnet(X,
                    y,
                    family = "multinomial",
                    alpha = alpha_grid[i],
                    nfolds = 5)

  cv_errors[i] <- min(cvfit$cvm)
  cv_models[[i]] <- cvfit
}

## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
## one
## multinomial or binomial class has fewer than 8 observations; dangerous ground

## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
## one
## multinomial or binomial class has fewer than 8 observations; dangerous ground

## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
## one
```

```

## multinomial or binomial class has fewer than 8 observations; dangerous gro
und
## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
one
## multinomial or binomial class has fewer than 8 observations; dangerous gro
und
## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
one
## multinomial or binomial class has fewer than 8 observations; dangerous gro
und
## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
one
## multinomial or binomial class has fewer than 8 observations; dangerous gro
und
## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
one
## multinomial or binomial class has fewer than 8 observations; dangerous gro
und
## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
one
## multinomial or binomial class has fewer than 8 observations; dangerous gro
und
## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
one
## multinomial or binomial class has fewer than 8 observations; dangerous gro
und
## Warning: from glmnet C++ code (error code -95); Convergence for 95th lambd
a
## value not reached after maxit=100000 iterations; solutions for larger lamb
das
## returned

## Warning in lognet(x, is.sparse, y, weights, offset, alpha, nob, nvars, :
one
## multinomial or binomial class has fewer than 8 observations; dangerous gro
und

cv_errors

## [1] 1.481258 1.457818 1.444237 1.488819 1.457805 1.474487 1.450907 1.4677
38
## [9] 1.482244 1.466520 1.468295

best_alpha <- alpha_grid[which.min(cv_errors)]
best_alpha

## [1] 0.2

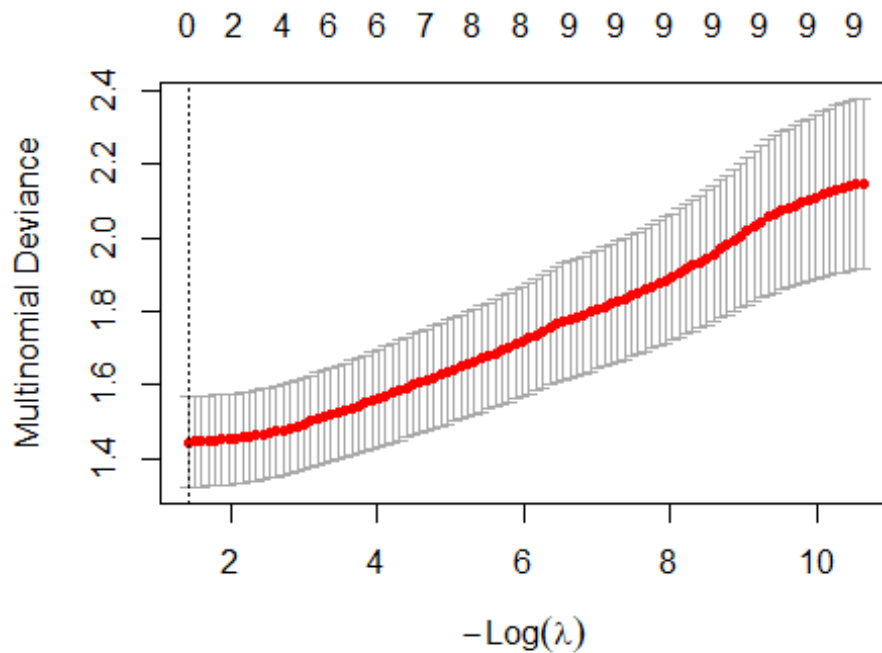
best_model <- cv_models[[which.min(cv_errors)]]

```

```
best_lambda <- best_model$lambda.min
best_lambda

## [1] 0.2416488

plot(best_model)
```



```
coef(best_model, s = "lambda.min")

## $Never
## 10 x 1 sparse Matrix of class "dgCMatrix"
##           lambda.min
## (Intercept)  1.883445
## SexMale      .
## Age          .
## ExerNone     .
## ExerSome     .
## SexMale:Age  .
## SexMale:ExerNone .
## SexMale:ExerSome .
## Age:ExerNone .
## Age:ExerSome .
##
## $Occas
## 10 x 1 sparse Matrix of class "dgCMatrix"
##           lambda.min
## (Intercept) -0.4085584
```

```

## SexMale      .
## Age          .
## ExerNone     .
## ExerSome     .
## SexMale:Age  .
## SexMale:ExerNone .
## SexMale:ExerSome .
## Age:ExerNone .
## Age:ExerSome .
##
## $Regul
## 10 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.min
## (Intercept)   -0.519784
## SexMale      .
## Age          .
## ExerNone     .
## ExerSome     .
## SexMale:Age  .
## SexMale:ExerNone .
## SexMale:ExerSome .
## Age:ExerNone .
## Age:ExerSome .
##
## $Heavy
## 10 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.min
## (Intercept)   -0.9551021
## SexMale      .
## Age          .
## ExerNone     .
## ExerSome     .
## SexMale:Age  .
## SexMale:ExerNone .
## SexMale:ExerSome .
## Age:ExerNone .
## Age:ExerSome .

```

The elastic net multinomial model included Sex, Age, Exer, and all two-way interactions as candidate predictors. Five-fold cross-validation was used to choose alpha and lambda. Alpha controls the balance between ridge and lasso penalties. Alpha = 0 is ridge, alpha = 1 is lasso, and values between 0 and 1 are elastic net.

The best alpha from the grid search was 0.2.

The best lambda was 0.2416488.

At lambda.min, all the coefficients were zero. This means the selected elastic net model didn't think anything was a useful predictors of Smoke. This means that the predictors did not make the cross-validated prediction good enough to overcome the penalty. The warning about some classes having fewer than 8 observations is important. It means some smoking categories, especially Hea

vy, have small sample sizes. This makes the multinomial elastic net model less stable.

#e. The hierarchy principle means that if an interaction term is included in a model, the related main effects must be included. In the selected elastic net model, none of the interaction terms were selected and no main effects were selected either. Since the model only kept intercepts, the hierarchy principle is not violated. If an interaction was picked without the main term, one way to fix it would be to use a hierarchical penalized regression method that forces main effects to stay in the model whenever their interaction terms are selected.